

Unit 2: Supervised Learning

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Introduction to Supervised Learning (SL)

What is Supervised Learning?

- Supervised learning is a subset of Artificial Intelligence and Machine Learning.
- It involves training algorithms to categorise data or predict outcomes by learning patterns from labelled datasets.
- In SL, the model is trained on labelled data i.e. each input is given to the model with the correct output.
- The main purpose of SL is to predict outcomes or classify information based on the patterns learnt from the past training.
- We use SL to teach the model using training data (i.e with correct answer) so that when new data is given to the model, it can classify accurately or predict outcomes correctly.
- EXAMPLE=You show a computer many pictures of animals and each picture is labelled as “cat”, “dog” and “bird”.

The computer learns patterns from these examples-like the shape of the ears, fur or size ,then it tries to predict the right label for new and unseen pictures.

Working

- **Training Data:** The model is provided with a training dataset that includes input data (features) and corresponding output data (labels or target variables).
- **Learning Process:** The model learns by comparing its predictions with the actual answers provided in the training data.
 - This is achieved by adjusting the model's parameters to minimize the difference between its predictions and the actual labels.
 - Over time, it adjusts itself to minimise errors and improve accuracy.

STEP-WISE WORKING-

1)Start with Labelled data- You begin by providing the dataset where each input has been paired with the correct output.

2)Split the data- The dataset is divided into:

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1. Training data= used to train the model .(80%)
2. Testing data= used later to check how well the model performs .
The splitting of data ensures that we are testing the model on data that it hasn't seen before.(20%)

3)Choose a Suitable algorithm- Based on the problem, you choose a SL algorithm,

For classification: SVM, KNN, Naïve Bayes, Decision Tree

For Regression: Linear regression, Ridge, Lasso

4)Learns Patterns from Training Data- The model starts analysing the input features and their corresponding output labels. It tries to learn how the inputs relate to the output. **EXAMPLE**=If the model is given a picture of a strawberry, then it will analyse the picture and learn the shape colour and size of the fruit and remember that!

5)Make Predictions and Compare- The model uses its current knowledge to make predictions on the training data. And compares its predictions with the corresponding output provided.

If wrong ,then **it calculates how far off** the prediction is.

6)Adjust to Reduce Errors-(Learning Process)= This is achieved by adjusting the model's parameters(**done by using gradient descent**) to minimize the difference between its predictions and the actual labels.

Over time, it adjusts itself to minimise errors and improve accuracy.

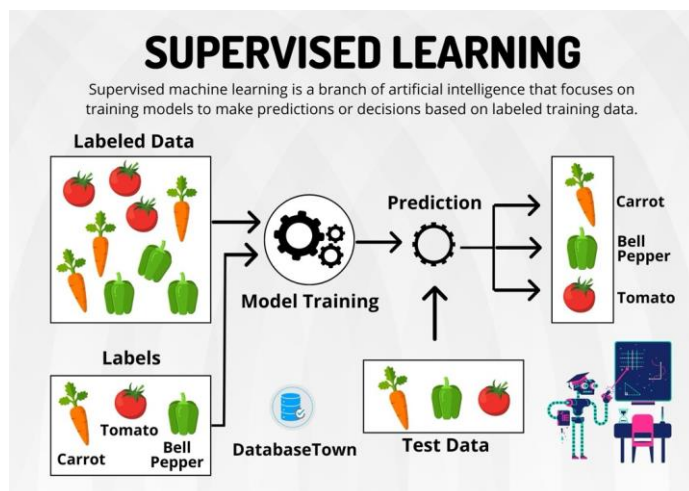
7)Repeat until Accuracy Improves- Steps 5 and 6 are repeated many times.

Each time , the model gets better at making correct predictions .**Eventually**, the model performs well on training data.

8)Test the Model- Now the model is tested using the **testing dataset** which is **unseen** by the dataset. This helps us to understand better how well it generalises to new data.

9)Evaluate Performance- We use metrics to measure the performance of the model:

1. **Accuracy:** Number of correct predictions.
2. **Precision and Recall:** Important in imbalanced datasets.
3. **F1 score:** Its the balance between precision and recall.
4. **Confusion Matrix:** Shows correct and incorrect predictions in detail.



Classification

- Classification is a subset of Supervised Learning.
- Classification teaches a machine to sort things into categories.
- The categorised data predefines classes based on patterns learned from training data.
- **GOAL**- To assign an input to a specific class or category.
- **EXAMPLE**- Email-spam or not , Transaction-fraud or not ,etc.
- There are two main types of classification.

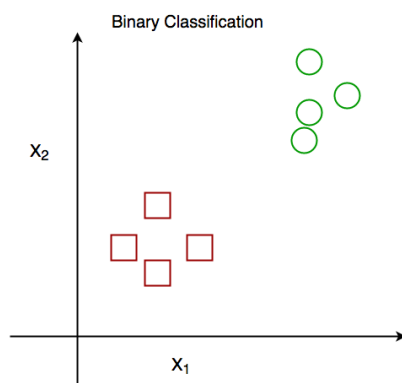
Types of Classification:

Binary Classification

- Simplest type of classification
- Data is classified into two distinct categories.
- **Training data is labelled in a binary format:** true and false (0 or 1) depending on the problem being tackled. The model learns the patterns that separate Class A from Class B.
- **EXAMPLE**- Medical diagnosis- whether a patient has diabetes (Yes/No) based on the test results.
- **1 means positive class (yes) and 0 means negative class (no).**

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- Other examples :Email detection , Loan-approved or rejected , etc.
- **Algorithms:**
 - Logistic Regression and Support Vector Machines
 - Other – K-Nearest Neighbours and Decision Tree

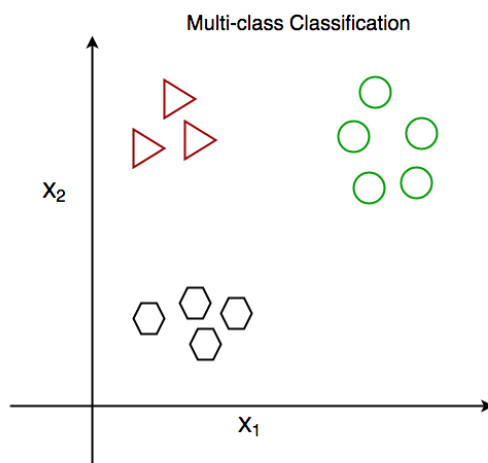


Multi-class classification

- Categorised into more than two classes.
- Selects one category from multiple possibilities
- **For example:** Image Recognition of animals (dog, cat, lion etc.)
- Examines features (like shape, colour and texture) to determine the correct class based on training data.
- It identifies patterns to decide to decide which one class the input fits best. Each input is tagged with only one label , but from multiple categories.

Like **LABEL={0=Cat,1=Dog, 2=Lion}**

- **Essential in real-world applications** where data belongs to one of many possible categories.
- **Other examples:** Handwriting recognition , Weather classification , Product category Prediction- electronics ,fashion and grocery , etc.



- **Algorithms:**

- Native – Random Forest, Naive Bayes, KNN, Gradient Boosting, SVM and Logistic Regression.
- For other algorithms, we must apply binary transformation approaches for multi-class tasks.
- These are **one-versus-one** and **one-versus-rest**.

One-versus-one:

- Trains a separate binary classifier for every possible pair of classes.
- For n classes, the number of classifiers needed is $N(N-1)/2$
- For each comparison, the classifier focuses only on two classes at a time.
- Final prediction is made by a majority vote. When a new input is given:
Each binary classifier votes for one of its two classes. The class which gets the most votes is the final predicted class.

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- Example: In a 3-class classification problem (Cat, Dog, Bird), the model trains three classifiers:
 - Classifier 1: Cat vs. Dog
 - Classifier 2: Cat vs. Bird
 - Classifier 3: Dog vs. Bird
- Best for SVM and Kernel-based algorithms because they work **efficiently with smaller datasets**.

One-versus rest:

- Also known as One-versus-all.
- Basically, each classifier is trained to differentiate one specific class from all other classes.
- For N classes, you train N classifiers. Each classifier learns: "Is this input from my class or not?"
- **Example:** If we have a 3-class classification problem (Cat, Dog, Bird), the model trains three classifiers:
 - **Classifier 1:** Cat vs. (Dog + Bird)
 - **Classifier 2:** Dog vs. (Cat + Bird)
 - **Classifier 3:** Bird vs. (Cat + Dog)
- When making predictions, model runs all N classifiers on a new input and assigns the label with the **highest confidence score / probability**.
- **Algorithms:** Logistic Regression, Decision Trees and Neural Networks

(Useful when classes are clearly separable but can be less accurate if they are alike or overlapping)

Multi-labelled Classification

- Multi label classification assigns multiple labels to an input.
- It allows each input to be in one or more than class at the same time.
- Unlike other classification models, where each instance is assigned a single label, multi-label classification allows for an instance may belong to multiple classes or none at all.

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- The model checks each label independently and predicts yes or no for every class.
- **Example:** in a text classification, a news article might be relevant to both 'sports' and 'economics' categories.
- **Common Algorithms used are:** Logistic Regression, KNN, Neural Networks and Decision Trees.

Imbalanced Classification

- Imbalanced classification occurs when the number of data points in different classes is unevenly distributed in the training dataset.
- Some classes have more data points, while others have very few.
- **This leads to bias in the model, favouring the majority class, misclassifying the minority class as noise leading to poor accuracy.**
- Examples: Fraudulent transaction detection(1 fraud transaction out of 10000), rare disease diagnosis, etc.
- Solutions:
 - Resampling techniques
 - SMOTE-It involves oversampling the minority class and undersampling the majority.
 - Class-weight adjustments . (more importance is given to minority class)
 - Anomaly detection methods

Classification Algorithms

Decision Tree

- Is an ML algorithm used for classification and regression tasks that forms tree-like structure with nodes (questions) and branches (decision/outcomes).
- It splits data into subsets based on the most significant feature at each step. Basically, data is split based on questions (feature values). Each node asks questions and branches lead to answers.
- The goal of a decision tree is to create a model that predicts the value of a target variable by learning simple decision rules inferred from the training data.
 - **Root Node:** is the starting point that represents the entire dataset.
 - **Branches:** These are the lines that connect nodes. It shows the flow from one decision to another.
 - **Internal Nodes:** are Points where decisions are made based on the input features.
 - **Leaf Nodes:** These are the terminal nodes at the end of branches that represent final outcomes or predictions

******Decision Tree uses Entropy and Information Gain (IG) to decide the best feature to split on:**

- **Entropy (H) :**
 - Entropy measures the impurity or randomness of a dataset.
 - It determines the uncertainty in a set of classifications.
 - Formula of entropy :

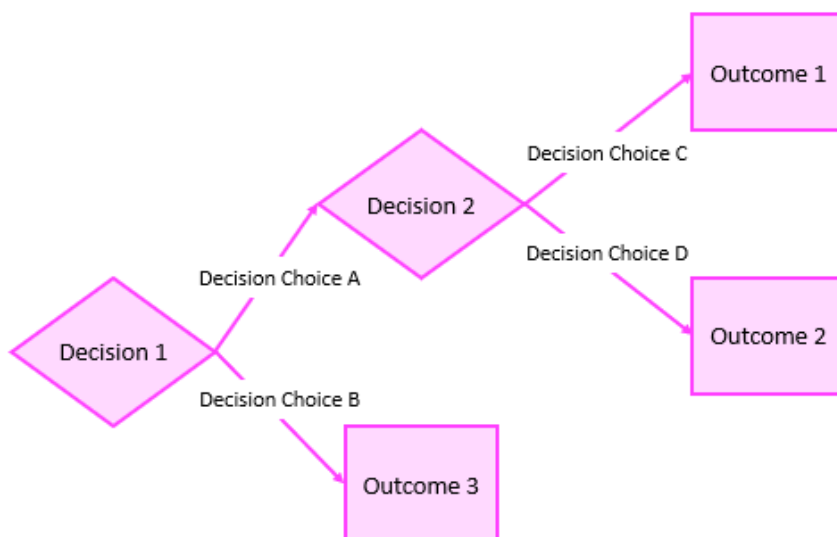
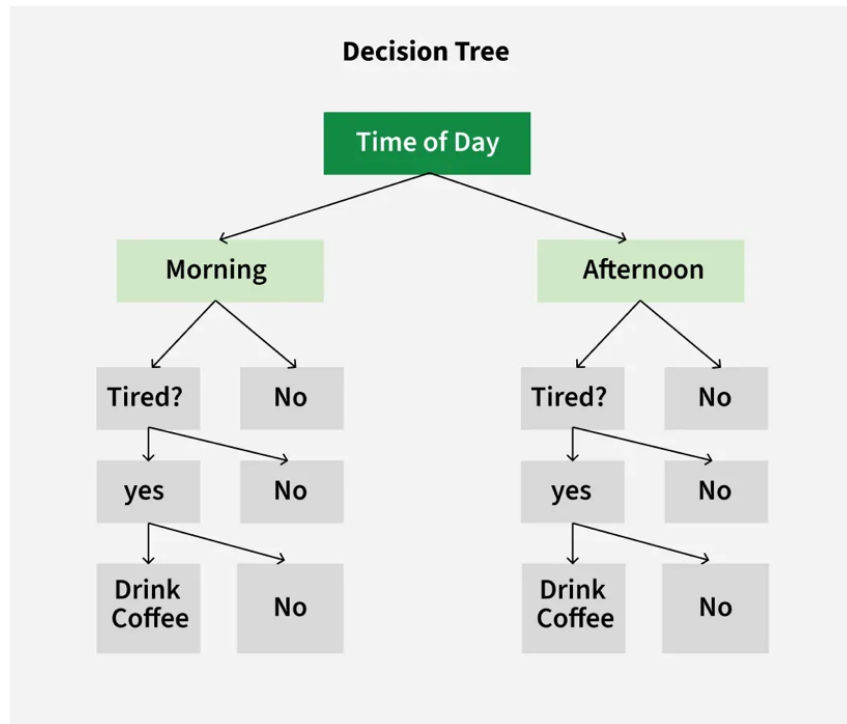
$$H(S) = - \sum p_i \log_2 p_i$$

Where,

- $H(S)$ is the entropy of dataset S
 - p_i is the probability of class i
 - $p_i \log_2$ represents the logarithmic base 2 of the probability
- **Information Gain (IG) :**
 - IG measures the reduction in entropy after splitting the dataset based on an attribute.
 - It helps in selecting the best attribute for decision trees.
 - Formula of IG :

$$IG(S, A) = H(S) - \sum \frac{|S_v|}{|S|} H(S_v)$$

- Where,
 - $IG(S, A)$ is the IG of attribute A
 - $H(S)$ is the entropy of original dataset
 - S_v is the subset after splitting by attribute A
 - $\frac{|S_v|}{|S|}$ is the proportion of samples in subset S_v
 - $H(S_v)$ is the entropy of the subset
- **Working:**
 - 1) Starting from the Root node=** A decision tree working starts with a main question known as the root node. This question is derived from the features that best splits the dataset. The is chosen using IG.
 - 2) Series of questions asked=** From the root node, the tree asks a series of yes/no questions.
 - 3) Data split:** Each question splits the data into subsets based on specific features.
 - For example:** if the first question is “Is it raining?”, the answer will determine which branch of the tree to follow.
 - This branching continues through a sequence of decisions.
 - 4) Repeat the process :** This step-by-step process continues until you have no more helpful questions.
 - 5) Reaching the Leaf node:** Finally, we reach the end of a branch (leaf node) where the final outcome or decision is found.
- **Advantages:**
 - Easy to understand and interpret
 - Can handle non-linear relationships
 - Doesn't require feature scaling
- **Disadvantages:**
 - Prone to overfitting (if too deep)
 - Prone to underfitting (if too shallow)
 - Can be unstable with small changes in data



Support Vector Machines

What is SVM?

SVM is powerful supervised learning algorithm mainly used for classification tasks. (also regression). It tries to draw boundary (hyperplane) that best separates different classes of the dataset.

- **Hyperplane:** A hyperplane is a decision boundary that separates different classes in a feature space. In linear classification, **it is represented by the equation $\mathbf{mx} + \mathbf{C} = 0$** . Data points on different sides of the hyperplane belong to different classes. It plays a crucial role in classification models like SVM.
- **Support Vectors (SVs):** Support vectors are the closest data points to the hyperplane. They are crucial in defining the optimal hyperplane and margin. Even if other data points change, support vectors determine the final decision boundary. Removing or modifying them can significantly impact classification accuracy. The model depends on these points to draw the separation line.
- **Margin:** The margin is the distance between the hyperplane and its nearest support vectors. SVM aims to maximize this margin to improve classification accuracy and generalization. A larger margin reduces the risk of overfitting. It ensures better separation between classes in feature space.
- **Kernel:** A kernel function transforms data into a higher-dimensional space to make it linearly separable. It helps SVM handle complex datasets where classes are not linearly divided. Popular kernels include linear, polynomial, and radial basis function (RBF), sigmoid etc. Choosing the right kernel improves model performance in classification tasks.
- **Performance Measure:** In machine learning, performance measures evaluate a model's effectiveness. Common metrics include accuracy, precision, recall, F1-score, and AUC-ROC. These metrics help assess classification, regression, and clustering models. The choice of metric depends on the problem type and dataset characteristics.

****AUC-ROC(Area Under Curve-Receiver Operating Characteristic)=It measures how well the model distinguishes between classes .Higher the AUC , better the models performance.**

- **Performance Parameters:** Performance parameters define how well a model performs. Key parameters include training time, inference time, memory usage, and generalization ability. For classification models, confusion matrix elements like true positives, false positives, and false negatives are crucial. Optimizing these parameters helps improve model efficiency and accuracy.
- **Kernel Functions:**
 - A kernel function helps the SVM deal with non linearly separable data by transforming it into a higher dimensional space, where it becomes easier to separate the classes using a HYPERPLANE . It helps support vector machines (SVM) and other models handle complex patterns. Kernels compute dot products in higher dimensions without explicit transformation. They enable efficient computation in high-dimensional spaces.
 - **Types of Kernel Functions:**
 - **Linear Kernel:** Suitable for linearly separable data.
 - **Polynomial Kernel:** Captures non-linear relationships using polynomial degrees.
 - **Radial Basis Function (RBF) Kernel:** Maps data to infinite dimensions, useful for complex decision boundaries.
 - **Sigmoid Kernel:** Like neural networks, useful for binary classification.
- SVs distinguish between two classes by finding the optimal hyperplane that maximizes the margin between the closest data points of opposite classes.
- The number of features in the input data determine if the hyperplane is a line in a 2-D space or a plane in a n-dimensional space.
- **Working:**
 - The key concept of the SVM algorithm is to find the hyperplane that best separates two classes by maximizing the margin between them.
 1. **Plot the data-** The model take labelled training data and plots it in space based on its features. Each point represents an example from a class(eg. spam or not spam).

2. **Find the best Hyperplane**- SVM tries to draw a boundary that best classifies the two classes .The hyperplane with the largest margin gives the most accurate classification result.
3. **Identify support vectors**- The data points closest to the hyperplane are called support vectors. These points are critical as they define the location of the hyperplane.
4. **Maximise the Margin**- SVM tries to maximise the margin—it is the distance between the hyperplane and the support vectors. Larger margin means the model is confident and can generalise better to unseen data.
5. **Use kernel**: If the data cannot be separated by a hyperplane ,then SVM uses kernel function to transform data into higher dimensions where it becomes linearly separable. Common kernels include: Linear, polynomial , RBF(gaussian) ,sigmoid.

- **Types of SVM:**

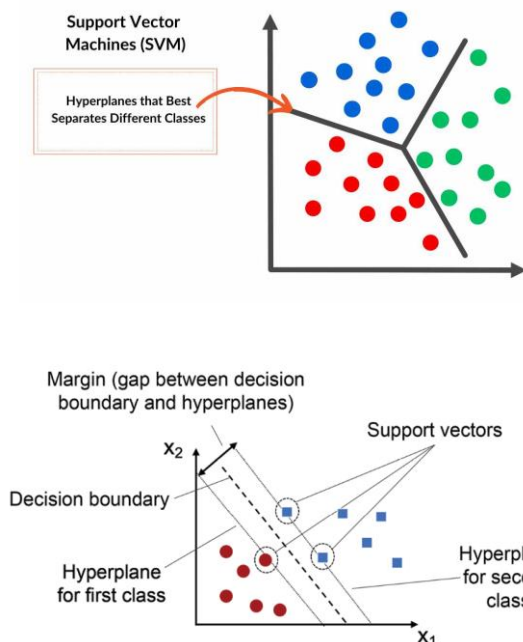
- Linear: Linearly separable data.
- Non-linear: Uses kernel functions for complex data.

- **Advantages:**

- Works well in high-dimensional spaces.
- Memory-efficient, using only support vectors.
- Effective for small-to-medium datasets.
- Effective when there is a clear margin of separation.

- **Disadvantages:**

- Computationally expensive for large datasets.
- Choosing the right kernel is tricky.
- Less effective when data is highly noisy or overlapping.
- Hard to interpret.



Random Forest

- Random forest is based on **ensemble learning method** i.e. it combines multiple decision trees to make predictions.
- Used for classification and regression.
- Each tree is trained on a random subset of the data and features.

For Classification- It uses majority voting.

For Regression- It takes average of the predictions.

- This approach helps improve the accuracy of predictions.
- **Working:**
 1. **Start with Full Dataset-** Includes the collection of examples you want to learn from(eg- emails, photos etc).
 2. **Create many Small Datasets using Bootstrapping-** The computer randomly selects rows (which might repeat) from the original dataset to form a new sample. This all is done with replacement which means that some rows might repeat while some might not appear at all from the dataset.

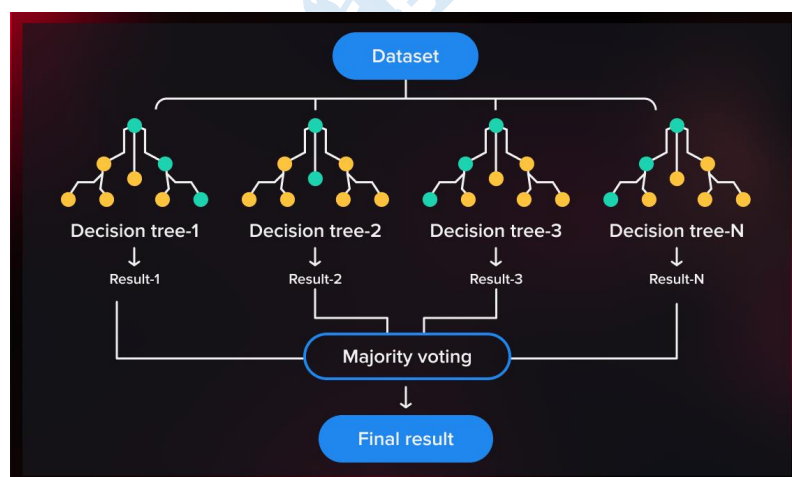
3. **Build many decision trees-** For each small dataset, the computer builds a decision tree. Each tree only sees some features, not all- this adds randomness and variety.
4. **Make predictions using all trees-** When new data comes in (like a new email), it is sent to every tree. Each tree gives its own prediction.
5. **Combine the results-** 1)**For Classification-** The trees vote and the most voted tree wins.
2)**For Regression-** Trees give numbers and the average is taken as the final result.

- **Advantages:**

- Reduces overfitting compared to single decision tree
- Works well with both classification and regression
- High accuracy and robustness

- **Disadvantages:**

- Harder to interpret
- Computationally expensive for large datasets
- Requires careful hyperparameter tuning



Naive Bayes

- NB is a probabilistic ML algorithm based on the Bayes' Theorem used for classification tasks.
- Bayes' Theorem- It calculates **how likely** something is to happen based on prior knowledge.
- It is termed 'Naive' because it assumes the features in the model are independent of each other. i.e. changing the value of one feature, does not directly influence any other features used in the algorithm.

EXAMPLE-In email classification, it assumes that the presence of the word "free" is independent of the word "offer", even if they appear together.

- Calculates the probability of a class given certain features.
- Bayes' Theorem finds the probability of an event occurring given the probability of another event that has already occurred.
- Bayes' theorem is stated mathematically as:

$$P(y|X) = \frac{P(X|y)P(y)}{P(X)}$$

- The Bayes' theorem assumes that the features are independent of each other. Hence, we split the evidence into independent parts.
- If X and y are two independent events, then:

$$P(y, X) = P(y)P(X)$$

- Let $(x_1, x_2, \dots, x_n)$ be a feature vector and y be the class label corresponding to this feature vector:

$$P(y|x_1, \dots, x_n) = \frac{P(y)P(x_1, \dots, x_n|y)}{P(x_1, \dots, x_n)}$$

Since $(x_1, x_2, \dots, x_n)$ are independent of each other:

$$P(y|x_1, \dots, x_n) = \frac{P(y) \prod_{i=1}^n (P(x_i|y))}{P(x_1, \dots, x_n)}$$

Since $P(x_1, x_2, \dots, x_n)$ in denominator remains constant for any given input, we can remove that term :

$$P(y|x_1, \dots, x_n) \propto P(y) \prod_{i=1}^n (P(x_i|y))$$

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To create a classifier model, compute the probability of a given set of inputs for all possible values of y and select the output with the highest probability.

This is mathematically expressed as:

$$\hat{y} = \underset{y}{\operatorname{arg\,max}} P(y) \prod_{i=1}^n (P(x_i|y))$$

In case of **Gaussian NB**, $P(\mathbf{x}_i | y)$ is calculated as:

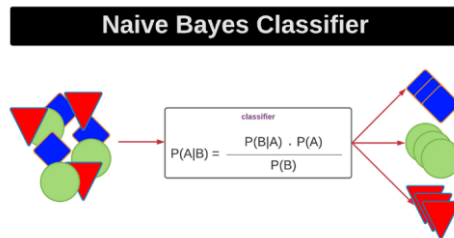
$$P(x_i|y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(x_i - \mu_y)^2}{2\sigma_y^2}\right)$$

How it works:

- For each class (eg- spam, not spam), it calculates the probability of that class.
- For each feature(eg specific words), it calculates the probability that the feature appears in each class.
- It combines these probabilities to find the most likely class for the input.
- **Types:**
 - **Gaussian Naive Bayes:** Continuous features follow a normal distribution.
 - **Multinomial Naive Bayes:** Used for text classification, handling word counts or frequencies.
 - **Bernoulli Naive Bayes:** Works with binary features.
- **Advantages:**
 - Easy to implement and computationally efficient.
 - Effective in cases with large number of features.
 - Performs well with limited training data.
- **Disadvantages:**
 - Assumes that features are independent.
 - Can be influenced by irrelevant attributes.
 - May assign zero probability to unseen events.
- **Applications:**
 - Spam email filtering

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- Sentiment analysis
- Medical Diagnosis
- Weather Prediction



K-Nearest Neighbours (KNN)

- It's the simplest classification algorithm.
- Used for classification and regression tasks.
- KNN is aka **lazy learner algorithm** because it does not learn from the training set immediately-It stores the data and classifies new data points based on similarity with nearby data points in the feature space..
- It does not depend on the structure of the data. It is a non-parametric algorithm.
- The basic idea is to predict the class of a new datapoint based on the majority class of its nearest neighbours in the training data.
- **Working:**
 - **Choose K:** Select the number of nearest neighbours (**K**) to consider. Choose the K wisely as smaller value of K is sensitive to noise while the larger value might ignore the details. **The most preferred value of K is 5.**
 - **Calculate Distance:** Compute the distance between the new data point and all existing data points. Metrics like **Euclidean distance(mostly used)**, **Manhattan distance**, or **Minkowski distance** are used.
 - **Find Nearest Neighbours:** Identify **K closest** data points.
 - **Final Prediction:**
 - **For Classification:** The algorithm looks at which category the neighbours belong to and picks the one that appears the most. This is called **majority voting**.

- **For Regression:** The algorithm looks for the K closest points and takes the **average** of the values of those K neighbours.

- **Advantages:**

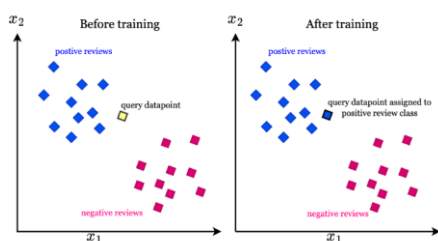
- Simple & Easy to Implement
- Can handle multi-class classification
- Works well with small datasets

- **Disadvantages:**

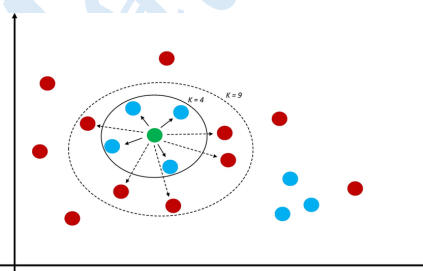
- **Computationally expensive** for large datasets.
- **Sensitive to outliers**
- **Choosing the right K value is crucial**

- **Applications:**

1. Spam email classification



2. Medical Diagnosis



Logistic Regression

- Used for binary classification.
- Logistic regression is a statistical algorithm which analyses the relationship between two data factors.
- The goal is to predict the probability that an instance belongs to a given class using the sigmoid function.
- It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, it gives the probabilistic values which lie between 0 and 1.
- In Logistic regression, instead of fitting a regression line, we fit an “S” shaped logistic function.

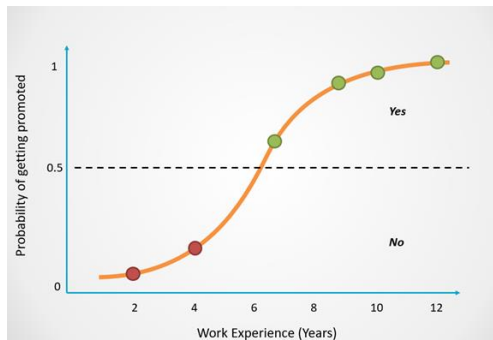
- Though it is used for classification, its called regression because it uses a regression like equation , but applies a sigmoid function to convert output into probabilities between range 0 and 1.
- The sigmoid function is a mathematical function used to map the predicted output values to probabilities. The S-form curve is called the **Sigmoid function** or the **logistic function**.
- **Working:**
 - 1) $\sigma(z) = \frac{1}{1+e^{-z}}$
 - 2) $z = w \cdot X + b$
 - Where w is weights or coefficient, and b is the bias term aka intercept.
 - the figure sigmoid function converts the continuous variable data into the probability i.e. between 0 and 1.
 - The first formula maps any value in the range of 0 to 1.

STEPWISE WORKING:

1. Takes input features like age , salary etc.
 2. Applies a linear combination using weights(refer the 2nd formula).
 3. Passes the result through the sigmoid function to get the probability.
 4. Compares this probability with the actual label and calculates error.
 5. Uses Gradient Descent to adjust weights and reduce the error.
 6. Repeats the same process until the model predicts with high accuracy.
- **Decision Rule:**
 - $\sigma(z)$ tends towards 1 as $z \rightarrow \infty$
 - $\sigma(z)$ tends towards 0 as $z \rightarrow -\infty$
 - $\sigma(z)$ is always bounded between 0 and 1
 - where the probability of being a class can be measured as:
 - $P(y = 1) = \sigma(z)$
 - $P(y = 0) = 1 - \sigma(z)$
 - **Cost function** (Log Loss) is used to measure the error in classification.

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- **Gradient Descent:** Optimisation algorithm used to minimise the loss function and update weights.
- **Regularisation (L1/L2):** Helps prevent overfitting by adding penalty terms to weights.
- **Types of Logistic Regression:**
 - **Binary Logistic Regression** – Two classes (e.g., pass/fail, spam/not spam).
 - **Multinomial Logistic Regression** – More than two classes (e.g., predicting weather: sunny, rainy, cloudy).
 - **Ordinal Logistic Regression** – Ordered classes (e.g., customer satisfaction ratings: low, medium, high)
- **Advantages of Logistic Regression:**
 - Simple and easy to interpret
 - Works well for linearly separable data
 - Fast training and inference
 - Outputs probability values, useful for decision-making
- **Disadvantages of Logistic Regression:**
 - Assumes a linear relationship between independent variables and log-odds
 - Not effective for complex, non-linear patterns
 - Struggles with imbalanced datasets
- **Applications:**
 - **Spam Detection**
 - **Medical Diagnosis**
 - **Fraud Detection**
 - **Customer Churn Prediction**



Regression

- Supervised learning technique.
- Used to model and analyse the relationship between a dependent variable (target) and one or more independent variables (features).
- Mainly used for predicting continuous values (e.g., house prices, temperature, stock prices).

Types of Regression Techniques

Below are the different regression techniques:

1. Linear Regression
2. Ridge Regression
3. Lasso Regression
4. Polynomial Regression
5. Bayesian Linear Regression

Regression Algorithms

1. Linear Regression

- Linear regression learns from the labelled datasets and maps the data points with most optimized linear functions which can be used for prediction on new datasets.
- It's used to model the relationship between a dependent variable (target) and one or more independent variables (features) using a straight-line equation.
- It predicts the continuous output variables based on the independent input variable.
- For example, to predict house price, various factor are considered, such as house age, distance from the main city, location, area and number of rooms, etc.
- Linear regression uses all these parameters to predict the house price by considering a linear relation between the features and price of the house.

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- Linear regression finds the best fit line that shows the relationship between input(X) and output(Y).
- Here, the features of the house become the independent variables to predict the dependent variable i.e. the price of the house.
- It predicts **continuous values** such as sales, stock prices, etc.

- **Working:**

- In linear regression, we assume that the features are independent of each other and there is some sort of linearity between the dependent and the independent variables.
- The primary objective of linear regression is to locate the best-fit line.
- **Best fit line equation:** a straight-line equation that represents the relationship between the dependent and independent variables.

$$y = mX + b$$

- **The slope of the line** (m) indicates how much the dependent variable changes for a unit change in the independent variable.
- The model finds this line by minimising the **Mean Squared Error (MSE)**
- The model uses the following equation to predict the independent variable.
- **Equation:**

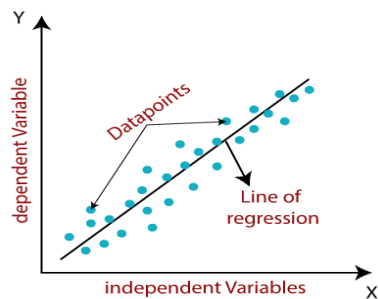
$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n$$

- Y = Predicted Output / Dependent Variable (Target)
- $X_1, X_2, \dots, X_n$ = Independent Variables (Features)
- b_0 = Intercept (bias)
- $b_1, b_2, \dots, b_n$ = Coefficients (weights)
- Finally, we find the best-fit line by finding the best values for b_0 (intercept) and b_1 (coefficient of x).
- In simple words, We collect the data that is input along with known output, then the data is plotted on the graph and linear regression finds the line that best fits all points (aka regression line).
- The model learns slope and intercept that minimises the error. And finally uses the line to predict Y for any new value of X .

Commented [SD1]: This means that changes in the independent variable lead to proportional changes in the dependent variable.

BML Unit 2: Supervised Learning

- The model uses a method **called Least Squares** to minimise the mean squared error (mse) that is it tries to reduce the distance between the actual and predicted values.
- **Advantages:**
 - Simple and easy to interpret.
 - Works well for linear relationships.
 - Computationally efficient for small datasets.
- **Disadvantages:**
 - Assumes a **linear relationship** (not suitable for non-linear data).
 - Sensitive to **outliers**.
 - Can overfit when using many correlated features (multicollinearity).
- **Uses:**
 - House price prediction
 - Stock market forecasting
 - Medical research (e.g., predicting cholesterol levels)



Difference between Classification and Regression

Feature	Classification	Regression
Definition	Predicts categorical labels	Predicts continuous values
Output Type	Discrete categories	Continuous values
Algorithms Used	Logistic Regression, Decision Tree, SVM, Random Forest, Neural Networks	Linear Regression, Lasso, Ridge, Polynomial Regression, Bayesian Regression
Example Use Cases	Email spam detection, Image recognition, Disease diagnosis	House price prediction, Stock market forecasting, Sales prediction
Performance Metrics	Accuracy, Precision, Recall, F1-score, ROC-AUC	Mean Squared Error (MSE), Root Mean Squared Error (RMSE), R^2 Score
Decision Boundary	Finds a boundary that separates categories	Finds a trendline/curve that fits data points
Handling Outliers	Less sensitive to outliers	Highly sensitive to outliers
Needs Feature Scaling	Sometimes (e.g., SVM, KNN)	Often needed (especially for gradient-based methods)

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